

**Table 5.1 Temporal Parameters of the H-Neuron**

| Parameter                          | Symbol        | Default Value                   | Description                                             |
|------------------------------------|---------------|---------------------------------|---------------------------------------------------------|
| Refractory Duration                | $R_t$         | 3 steps                         | Number of time steps spent in the REFRACTORY state      |
| Weak-Signal Accumulation Threshold | $W_{th}$      | 3                               | Number of weak inputs required to complete accumulation |
| State Cycle Length                 | $T_{cycle}$   | 4 steps                         | IDLE→ACTIVE→REFRACTORY→READY cycle                      |
| Simulation Time Step               | $\Delta t$    | $\approx 1$ ns                  | Adjustable simulation parameter                         |
| Strong-Signal Condition            | $S_{strong}$  | $z_a + z_b = 0$                 | Antiphase coupling criterion                            |
| Learning Trigger Condition         | $L_{trigger}$ | $z_{top} =$<br>$z_{bottom} = i$ | Topological reconstruction criterion                    |

**Weak-Signal Accumulation Parameters**

Weak signals do not directly induce learning.

Instead, they are accumulated through a three-stage process:

**First Weak Input**

$$z_{top} : 0 \rightarrow i$$

**Second Weak Input**

$$\theta_{core} \rightarrow \theta_{core} + \pi$$

**Third Weak Input**

$$z_{bottom} : 0 \rightarrow i$$

The accumulation counter is:

$$W = 0, 1, 2, 3$$

When:

$$W = W_{th} = 3$$

the system satisfies:

$$z_{top} = z_{bottom} = i$$

and becomes eligible for topological reconstruction.

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## Learning Criterion

Learning occurs only after accumulation is completed.

$$z_{top} = i$$

$$z_{bottom} = i$$

$$\Rightarrow$$

$$R_1 \leftrightarrow R_3$$

This topological reconstruction converts:

$$Weak \rightarrow Strong$$

connections.

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## Refractory Dynamics

After activation:

$$Rt = 3$$

the neuron enters the REFRACTORY state.

For:

$$0 < Rt \leq 3$$

the neuron:

- rejects all inputs;
- disables accumulation;
- disables learning;
- disables output.

The refractory counter evolves as:

$$Rt(t+1) = Rt(t) - 1$$

When:

$$Rt = 0$$

the neuron transitions into the READY state.